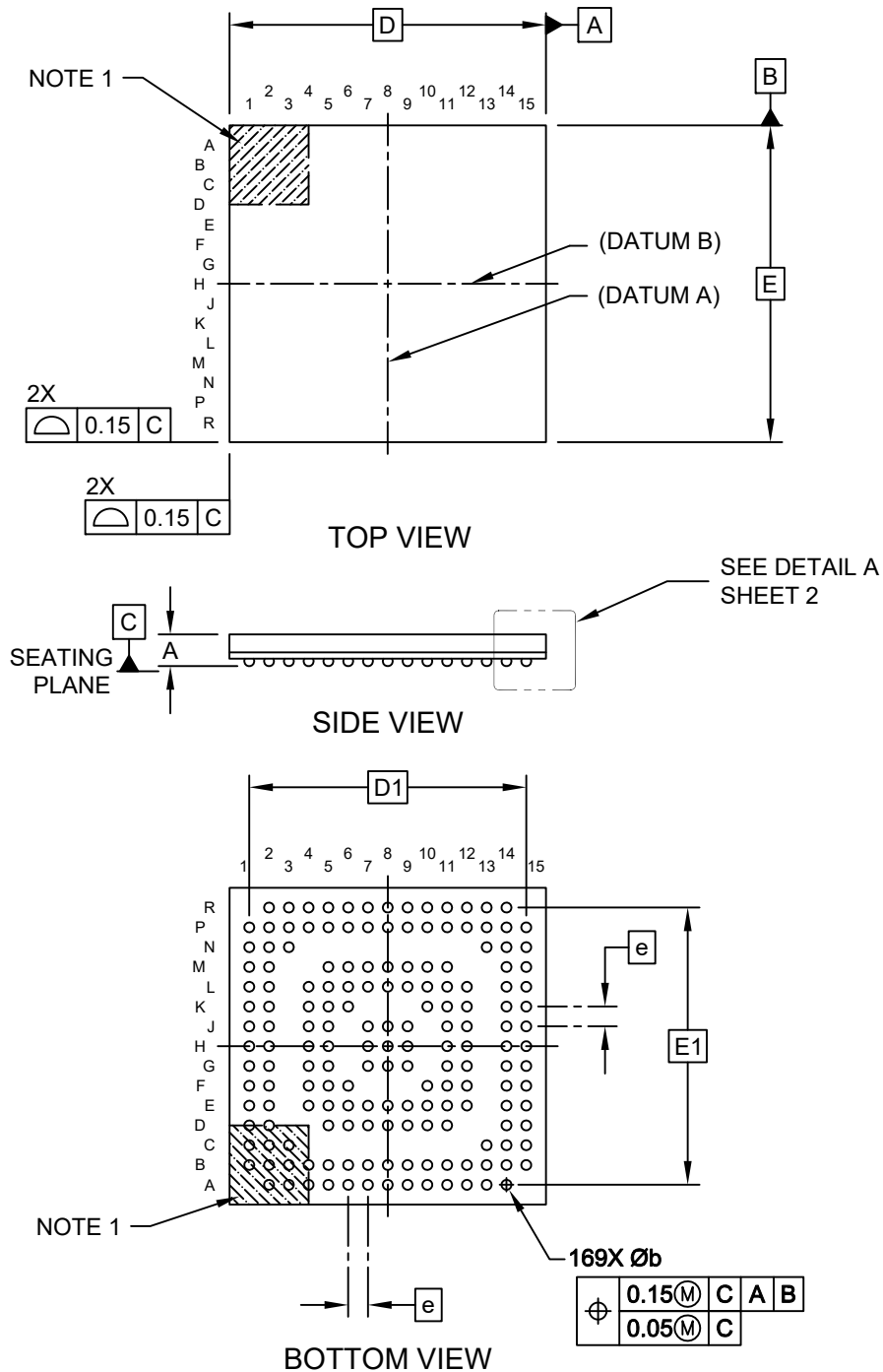


169-Ball Very Very Thin Fine Pitch Ball Grid Array (XYX) - 8x8 mm Body [WFBGA]

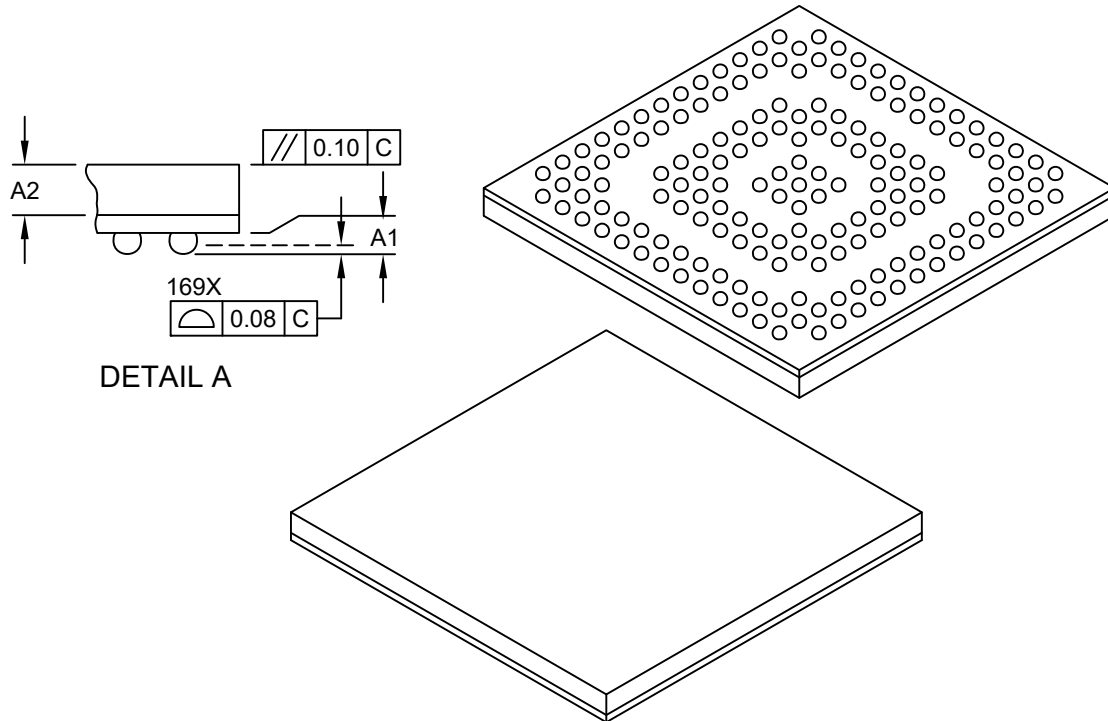
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Microchip Technology Drawing C04-470 Rev. A Sheet 1 of 2

169-Ball Very Very Thin Fine Pitch Ball Grid Array (XYX) - 8x8 mm Body [WFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



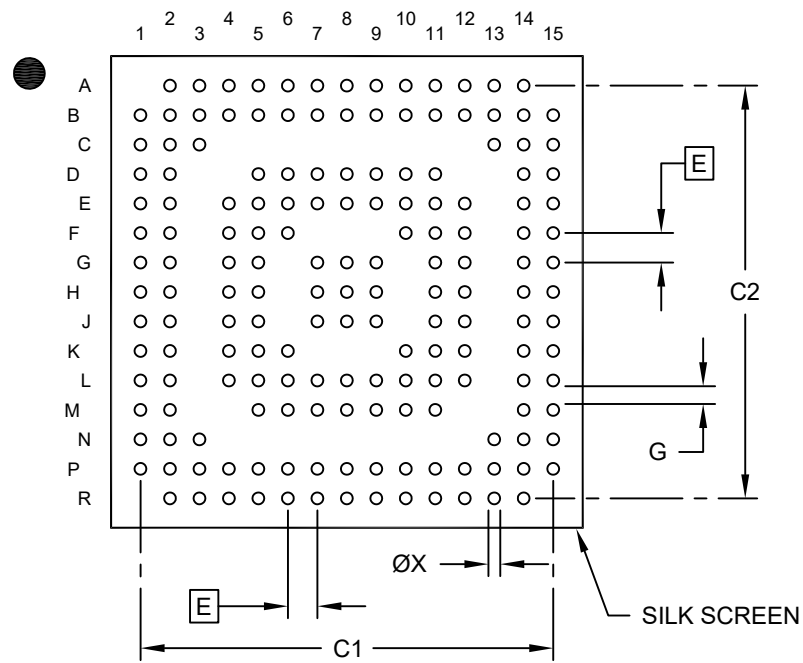
Dimension Limits	Units	MILLIMETERS		
		MIN	NOM	MAX
Number of Terminals	N	169		
Pitch	e	0.50 BSC		
Overall Height	A	-	-	0.80
Standoff	A1	0.14	0.19	0.24
Mold Thickness	A2	0.35	0.40	0.45
Overall Length	D	8.00 BSC		
Overall Terminal Spacing	D1	7.00 BSC		
Overall Width	E	8.00 BSC		
Overall Terminal Spacing	E1	7.00 BSC		
Terminal Diameter	b	0.20	0.25	0.30

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

169-Ball Very Very Thin Fine Pitch Ball Grid Array (XYX) - 8x8 mm Body [WFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Overall Contact Pad Spacing	C1		7.00	
Overall Contact Pad Spacing	C2		7.00	
Contact Pad Width (X169)	X			0.20
Contact Pad to Contact Pad	G	0.20		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-2470 Rev. A